PLATFORM FOR AUTONOMOUS INDOOR NAVIGATION USING SLAM, META-RL, AND MULTI-ROBOT COLLABORATION

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BSc (Hons) Degree in Information Technology

Specializing in Software Engineering

Department of Software Engineering

Sri Lankan Institute of Information Technology Sri Lanka

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ABSTRACT

Autonomous navigation in dynamic and unpredictable environments requires systems that can adapt their decision-making in real time while maintaining accurate environmental awareness. The approach outlined in this work combines Simultaneous Localization and Mapping with Meta-Reinforcement Learning (Meta-RL) in a decentralized yet cloud-assisted architecture, enabling multi-robot teams to operate efficiently in changing conditions. The design focuses on continuous policy adaptation, distributed data processing, and human-in-the-loop control to enhance scalability and operational resilience. At the edge layer, autonomous robots equipped with LiDAR and camera systems gather detailed environmental data while navigating. Each robot processes sensor inputs locally for immediate obstacle avoidance and short-term path planning. Data is then transmitted to a local edge server, which merges information from multiple robots, filters noise, and compiles preliminary maps. This step reduces the computational and bandwidth burden on the robots themselves, ensuring low-latency responses in time-sensitive scenarios. The cloud processing layer manages global computation tasks, including high-fidelity SLAM processing, map fusion across all deployed robots, and centralized policy training through Meta-RL. By integrating multi-robot data into a unified 3D map, the system gains a shared situational understanding, which the Meta-RL module uses to refine navigation strategies. The cloud environment continuously trains and updates navigation models, pushing optimized policies back to the edge devices without requiring manual reprogramming. This loop allows the robots to learn from both past experiences and newly encountered environments. A dedicated web-based visualization interface gives operators real-time access to robot locations, sensor data, and environmental maps. This interface supports live monitoring as well as human feedback input, which is incorporated into future policy adjustments. This design ensures that domain expertise can be combined with automated learning, leading to more robust and context-aware navigation.

Keywords - Autonomous Navigation, Multi-Robot Systems, SLAM, Meta-Reinforcement, Edge-Cloud Architecture

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AMR	Autonomous Mobile Robot
AGV	Automated Guided Vehicle
SLAM	Simultaneous Localization and Mapping
Meta-RL	Meta-Reinforcement Learning
MAML	Model-Agnostic Meta-Learning
LiDAR	Light Detection and Ranging
RGB-D	Red, Green, Blue – Depth
IMU	Inertial Measurement Unit
CORB-SLAM	Collaborative ORB-SLAM
DDF-SLAM	Distributed Data Fusion SLAM
GPU	Graphics Processing Unit
TPU	Tensor Processing Unit
DRL	Deep Reinforcement Learning
3D	Three-Dimensional
UAV	Unmanned Aerial Vehicle
API	Application Programming Interface
IoT	Internet of Things

1 INTRODUCTION

Autonomous indoor navigation in unknown and dynamic environments has long been a challenge in robotics, but it is becoming increasingly important in areas such as smart homes, automated warehouses and disaster recovery scenarios. The ability of robots to travel successfully without prior maps is significantly relied on SLAM, a technique that allows for real time environment reconstruction while tracking the robot's position [1]. SLAM has made significant advances with optical, LiDAR and multi sensor fusion techniques providing robust performance in controlled environments. However most traditional SLAM algorithms assume static or semi static surroundings and a single robot operational model, limiting their effectiveness in real world applications where settings are fundamentally dynamic [4].

Dynamic components including moving individuals, moving furniture, clear surfaces and reflective materials pose significant problems to typical SLAM systems. These issues appear as false loop closures, map inconsistencies, and loss of localization accuracy which could risk navigation safety and efficiency. Rapid adaptation is crucial in highly dynamic environments, yet standard rule-based path planning and control systems frequently fail to respond successfully [5].

Meta-RL has emerged as a potential method for rapid robotic adaptability. Unlike the traditional reinforcement learning methods, Meta-RL allows systems to make assumptions from previous experiences and quickly modify navigation strategies to the new unknown environments [10]. Meta-RL has demonstrated great potential for single robot navigation but how it can be used in a multi-robot collaborative context is mostly unexplored due to various reasons. This gap is significant since collaborative navigation offers various advantages such as simultaneous exploration for efficient mapping, shared situational awareness and resilience through distributed decision-making [6].

The multi-robot collaboration framework is a key component for these advantages. It allows real time map sharing, task allocation, conflict resolution and synchronized exploration. By merging local maps into a consistent global map [8], robots can leverage one another's discoveries. Our framework operationalizes these requirements

with decentralized coordination and robust map fusion and scheduling mechanisms, designed for heterogeneous robot teams [7].

This proposal introduces an integrated platform for Autonomous Indoor Navigation by combining advanced SLAM with Meta-RL based adaptation engine, a decentralized multi robot collaborative framework and a 3D mapping and visualization platform. The system utilizes an edge-cloud architecture, in which lightweight robots with simple sensors transporting heavy computations like SLAM processing, map fusion, and policy optimization to a centralized server. This design offers scalability, cost-efficiency, and interoperability with a variety of robot types while preserving real-time performance [6].

From a software engineering approach, the platform encourages flexibility, reusability, and scalability to facilitate the integration of emerging algorithms, sensors, and robotic platforms. Along with SLAM, Meta-RL, and the multi-robot cooperation architecture, the 3D Mapping & Visualization Platform offers an easy-to-use interface for real-time monitoring and debugging for operators. These components improve autonomous robotics by allowing multi-robot teams to move collaboratively, adapt to dynamic changes, and keep consistent global maps in complicated interior environments. The suggested solution bridges the gap between academic research and industry implementation, increasing logistical efficiency, ensuring safety in hazardous areas, and allowing new robotics applications in human-centric environments.

1.1 Background and Literature Survey

The evolution of autonomous systems has altered the way indoor logistics, inspection and service activities are carried out. Historically, tasks like item transportation and inventory management relied significantly on human labor which was inefficient, physically demanding and error prone. Automated Guided Vehicles (AGVs) were the first wave of automation to enable repeated activities in organized and controlled environments. However, AGVs depended on permanent navigation infrastructure, such as magnetic strips or painted floor lines, rendering them inefficient in the places with layout changes and dynamic obstacles [1].

The introduction of Autonomous Mobile Robots (AMRs) marked a significant leap forward. AMRs can dynamically detect and traverse new surroundings by utilizing advanced sensing methods such as LiDAR, RGB-D cameras and inertial measurement units (IMUs) as well as SLAM algorithms [3]. This enables its usage in complicated, real-world environments such as warehouses, hospitals and smart buildings. However, in large-scale applications a single AMR is insufficient. The true potential lies in deploying fleets of robots that can work together to explore and operate more effectively. This transition from a single agent to multi agent system creates a new set of challenges that define the boundary of current robotics research [7].

Indoor environments are hardly static because moving people, rearranging furniture and temporary obstacles cause a shift in distribution between training and operating settings. While Deep Reinforcement Learning (DRL) has shown ordinary DRL agents require considerable training in new settings, making them not suitable for applications that require immediate attention. Meta-RL solves this problem by allowing agents to "learn how to learn" and change rules with little new knowledge [10]. For example, Google research conducted in 2020 demonstrated fast adaptability in legged robots through evolutionary meta-learning [11]. However, developing whole lifecycle platforms from simulation-based training to real world deployments and constant fine-tuning is still an unexplored topic.

Scaling from a single robot to a collaborative team raises serious distributed system challenges. To outperform a single agent robot, a team of robots must coordinate their activities to achieve a shared goal, such as exploring an unexplored region. This requires addressing several key challenges:

- **Distributed Model Training and Deployment**: In multi-robot systems, efficiently training and deploying machine learning models across the team presents additional challenges. Models may need to be updated based on collective data from all robots, requiring synchronization and bandwidth-efficient distribution. Ensuring that each robot runs the latest optimized model while minimizing communication overhead and latency is crucial for maintaining coordinated behavior in real time.

- **Communication and Data Fusion**: Agents must effectively share sensor data and local maps with each other. Combining these incomplete maps into a single global map is a difficult data fusion task [6]. Systems such as CORB-SLAM have created frameworks for this but they frequently rely on protocols that may struggle with scalability and network instability.
- **Dynamic Task Allocation**: Dynamic task allocation is appointing different agents for different areas of a building to explore in order to increase coverage and reduce redundancy. Cooperative navigation techniques are vital for efficiency but building decentralized algorithms that lead to globally optimum behavior remains a persistent difficulty [8].
- **Conflict Resolution**: Managing path conflicts and right-of-way to prevent collisions and deadlocks in shared environments.

These challenges require robust communication protocols, consensus algorithms, and lightweight yet scalable data-sharing strategies.

A multi-robot fleet generates a continuous stream of high-dimensional real-time data which includes raw sensor feeds, SLAM-generated maps, navigation choices, job allocations and inter-robot communication. Without the appropriate tools, this data can become overwhelming for operators, which can make the system impossible to analyze, diagnose and optimize. Existing interfaces often provide only basic 2D positional data, lacking advanced features such as collision heatmaps, historical performance analytics, and interactive map annotations [12].

The absence of integrated visualization and diagnostic tools in many existing multi-robot frameworks is a major obstacle to scaling. Without them, troubleshooting involves manual log analysis, which is time-consuming and prone to error.

To address this difficulty, a 3D Mapping & Visualization Platform must be developed that not only depicts real-time global maps and robot data, but also includes historical analytics, performance heatmaps, and interactive controls. Such a system converts raw sensor and navigation data into usable insights, allowing operators to make quick decisions, save downtime, and retain confidence in the system's performance even if the number of robots and mission complexity increases.

Table 1.1 Core Challenges in Multi-Robot System Deployment

Factor	Practical Effect on Robots	Typical Symptom	Impact (1-5)
Distributed Model Training and Deployment	Delays or inconsistencies in model updates across robots; inefficient use of cloud resources	Robots act inconsistently and outdated models lead to poor coordination or decision-making	5
Real-Time Adaptation and Dynamic Obstacles	Inability to adjust to new or moving obstacles	Collisions, inefficient pathing, operational freezes	5
Multi-Robot Coordination and Scalability	Robots performing redundant work or interfering with each other	Inefficient area coverage, inconsistent global maps, bottlenecks	5
System Complexity and Operational Oversight	Inability for human operators to monitor, debug, or manage the fleet	Prolonged downtime, difficulty diagnosing failures, lack of trust in the system	4

The available literature typically addresses these issues in isolation. Meta-RL research may produce a highly adaptable agent, but it fails to investigate how it operates in multi-agent systems. Working on collaborative SLAM may help with map merging, but it assumes robots with basic, non-adaptive behaviors. The key gap therefore is the lack of a single, integrated platform to address these concerns comprehensively. Without a strong framework to regulate these interactions, a team of robots may be less efficient than a single agent. This difficulty is essentially a distributed systems engineering problem, requiring solutions that assure data consistency and allow for dynamic resource allocation across independent agents.

1.2 Research Gap

Autonomous navigation in indoor and semi-structured environments is a rapidly evolving field that combines robotic perception, planning, and control. Several research areas contribute significantly, including SLAM, reinforcement learning, multi-robot collaboration, and dynamic change detection. While individual studies demonstrate strong progress, a detailed comparative analysis reveals persistent gaps, particularly when considering real-time adaptability, multi-robot coordination, semantic-aware change detection, and full integration into a unified operational platform.

SLAM is fundamental for autonomous navigation, providing robots with the ability to localize and map unknown environments simultaneously. ORB-SLAM [1] introduced an efficient monocular SLAM system with high-accuracy localization and loop closure capabilities. While ORB-SLAM performs well in static environments, it struggles in dynamic scenarios with moving obstacles, limiting its applicability for real-world operations. RTAB-Map [2] extended SLAM to handle real-time RGB-D data and appearance-based loop closure. However, it is optimized for single-robot operations, and centralized processing limits scalability in multi-robot deployments. Cartographer [3] offers real-time 2D/3D mapping using LiDAR but has minimal dynamic obstacle handling and high computational requirements for multi-sensor fusion.

While these systems advance mapping accuracy, they do not integrate real-time adaptive decision-making, semantic awareness, or multi-robot collaboration, which are crucial for long-term autonomous operations in evolving environments. Reinforcement learning enables robots to learn optimal navigation strategies. Meta-reinforcement learning, particularly MAML [16], allows robots to rapidly adapt policies to unseen environments. Meta-RL for obstacle avoidance in simulated scenarios [17], showing improvements over standard RL. Despite this, most research remains simulation-based, lacking integration with real-world SLAM, real-time sensor feedback, and multi-robot coordination. Existing RL approaches also fail to handle sensor noise, communication delays, and the constraints of real-time control. Consequently, there is a critical gap in developing Meta-RL strategies that operate

seamlessly within a real-world SLAM-based, multi-robot system, enabling dynamic obstacle-aware navigation. Ulti-robot systems can significantly improve exploration efficiency and coverage. Distributed SLAM frameworks like CORB-SLAM [18] and DDF-SLAM [19] support map fusion and collaborative data sharing among multiple robots. These methods enhance global map consistency but rely heavily on network reliability, assume homogeneous robot teams, and often use centralized control. Task allocation and conflict resolution are typically rigid or static. Furthermore, most frameworks lack integration with adaptive learning policies or dynamic obstacle-aware navigation**,** reducing their utility in dynamic real-world scenarios.

Dynamic environments necessitate robust change detection to maintain accurate maps. Focus on methods for detecting structural [21] changes using multi-sensor [20] fusion and semantic labeling. While effective, these approaches are mostly offline or single-robot [21] focused and computationally intensive, limiting real-time deployment. There is no scalable system that supports real-time, decentralized semantic-aware map updates, particularly for multi-robot teams that need consistent and coordinated navigation in evolving indoor environments. Existing research demonstrates significant progress in isolated domains but fails to provide a unified solution. SLAM, Meta-RL, multi-robot collaboration, and dynamic change detection largely operate in isolation. No study fully integrates these capabilities into a comprehensive autonomous navigation platform that supports:

- Real-time adaptive navigation via Meta-RL,
- Decentralized multi-robot collaboration with dynamic task allocation and conflict resolution,
- Continuous 3D SLAM mapping with semantic awareness, and
- Real-time, version-controlled change detection integrated into operational workflows.

Bridging these gaps is critical for practical applications such as warehouse automation, indoor monitoring, and disaster response, where robots must adaptively navigate, collaborate, and update shared maps in real-time.

Table 1.2 Research Gap Analysis

Research Study	Focus Area	Strengths	Limitations / Gap	Relation to Proposed Comprehensive Platform	Reference
ORB-SLAM	3D SLAM	Accurate localization, loop closure	Static environments single robot lacks dynamic adaptation	Forms SLAM backbone; extended for multi-robot and dynamic updates	[1]
RTAB-Map	Real-time RGB-D SLAM	Loop closure; real-time mapping	Limited scalability; single-robot focus; centralized	Used as backend for collaborative multi-robot SLAM	[2]
Cartographer	LiDAR SLAM	Real-time 2D/3D mapping	High computation and minimal dynamic obstacle handling	Integrated into hybrid platform with adaptive obstacle awareness	[3]
MAML	Meta-RL	Rapid generalization to unseen tasks	Simulation only; lacks real-world integration; single robot	Provides Meta-RL adaptive navigation engine	[16]
Obstacle Avoidance	Meta-RL obstacle avoidance	Adaptive policies in simulation	Single robot; no multi-robot coordination and ignores sensor noise	Basis for real-time Meta-RL in multi-robot setup	[17]
CORB-SLAM	Multi-robot distributed SLAM	Map fusion; distributed architecture	Network-dependent; centralized task allocation; no adaptive policies	Inspiration for decentralized multi-robot collaboration framework	[18]
DDF-SLAM	Multi-robot SLAM	Data sharing; distributed mapping	Homogeneous robots; limited dynamic adaptation	Adapted for heterogeneous robot teams with adaptive path planning	[19]

From the comparative analysis, the key gaps are:

1. Real-Time Adaptive Navigation Gap**:** Most Meta-RL and RL approaches are simulation-based and lack real-world SLAM integration, limiting adaptive obstacle-aware navigation [17].
2. Multi-Robot Coordination Gap**:** Existing multi-robot SLAM systems depend on centralized control, homogeneous robots, and static task allocation, without integration of adaptive learning [19].
3. Dynamic Change Detection Gap**:** Semantic-aware change detection approaches are offline, single-robot focused, and computationally expensive [20].
4. Integration Gap: No research integrates SLAM, Meta-RL, multi-robot collaboration, and semantic-aware change detection into a comprehensive autonomous navigation platform**.**

The gaps clearly justify the need for a holistic platform that unifies these capabilities. The proposed system addresses these gaps by providing real-time adaptive navigation, decentralized multi-robot collaboration, continuous 3D SLAM mapping, and semantic-aware change detection. This comprehensive platform ensures robustness, scalability, and operational efficiency in unknown or evolving indoor environments, overcoming the limitations of existing research.

1.3 Research Problem

The rapid advancement of autonomous mobile robots (AMRs) has significantly improved indoor navigation capabilities, yet numerous challenges remain that hinder their deployment in dynamic and unstructured environments. Traditional Simultaneous Localization and Mapping (SLAM) techniques, such as ORB-SLAM [1] and RTAB-Map [2], have demonstrated high-accuracy mapping and localization in static or structured settings; however, their performance significantly declines when faced with dynamic obstacles, moving furniture, or temporary environmental changes.

Similarly, LiDAR-based mapping systems like Cartographer [3] provide real-time 2D and 3D mapping but are computationally intensive and lack the ability to adapt effectively to real-time dynamic conditions or integrate semantic awareness. While these SLAM systems form the backbone of robotic perception, they typically operate in isolation and do not natively support decentralized multi-robot collaboration or adaptive navigation strategies. In parallel, reinforcement learning approaches, particularly Meta-Reinforcement Learning (Meta-RL) techniques like MAML [16], have enabled rapid adaptation to unseen scenarios, allowing robots to generalize navigation policies to new environments.

Studies such as Zeng et al. [17] demonstrate the potential of Meta-RL for obstacle avoidance in simulated environments. However, these methods are primarily simulation-based, often ignoring real-world constraints like sensor noise, communication delays, and the latency inherent in real-time control. Furthermore, existing Meta-RL implementations rarely consider multi-robot environments where coordination, task allocation, and map-sharing are critical. Multi-robot collaboration frameworks, including CORB-SLAM [18] and DDF-SLAM [19], offer distributed SLAM solutions with map fusion and shared environmental awareness, enhancing global consistency and exploration speed. Nonetheless, these systems rely on centralized task allocation, assume homogeneous robot teams, and lack integration with adaptive Meta-RL policies for real-time obstacle-aware navigation, limiting their scalability and robustness in heterogeneous or large-scale deployments. Another persistent challenge is dynamic change detection. Furthermore, multi-sensor fusion [20]and semantic information to detect environmental changes [21] over time, improving map accuracy and navigation safety. Yet, most existing methods are offline or designed for single-robot scenarios, failing to support real-time, decentralized semantic-aware updates in multi-robot teams.

As a result, robots often operate with outdated maps, leading to inefficiencies and potential collisions. Collectively, these limitations highlight the lack of a unified solution that seamlessly integrates SLAM, Meta-RL adaptive navigation, multi-robot collaboration, and dynamic change [20] detection into a comprehensive autonomous

navigation platform. Such a platform is essential for enabling AMRs to navigate unknown indoor environments efficiently, adapt to dynamic obstacles in real time [21], share mapping and exploration data among heterogeneous teams, and maintain continuously updated semantic-aware maps. Addressing this problem requires a holistic approach that combines scalable software architectures, hybrid edge-cloud processing, robust communication protocols, and modular algorithmic pipelines. By integrating these capabilities, the proposed platform aims to provide an end-to-end solution that ensures real-time adaptability, operational robustness, and scalability, bridging the gap between isolated research contributions and practical deployment in real-world dynamic environments. This research problem directly motivates the design and implementation of a comprehensive autonomous navigation system capable of synergizing 3D SLAM mapping, Meta-RL-based obstacle-aware adaptation, decentralized multi-robot coordination, and semantic-aware change detection, thereby addressing the critical limitations present in the current state of the art.

2 RESEARCH OBJECTIVES

2.1 Main Objectives

The primary aim of this research is to design, develop, and validate a comprehensive autonomous navigation platform capable of operating efficiently in dynamic and unknown indoor environments. The platform will seamlessly integrate adaptive navigation algorithms, robust mapping systems, and multi-robot collaboration strategies into a unified ecosystem. This will involve the combination of advanced SLAM-based mapping, adaptive obstacle avoidance using reinforcement learning techniques, and a decentralized coordination mechanism to ensure scalability and robustness in multi-agent deployments. Furthermore, the platform will incorporate semantic change detection to maintain an up-to-date understanding of the environment, enabling the robots to make informed navigation and task allocation decisions in real time. The system will not only exist in simulation but will also be implemented and tested on a functional prototype robot, bridging the gap between theory and practical application. In addition to core navigation functionalities, the research aims to establish a strong model training pipeline and deployment strategy that allows rapid iteration and improvement of algorithms while ensuring they meet real-world operational constraints.Real-time monitoring tools will be integrated into the system to continuously assess model performance, detect anomalies, and trigger adaptive retraining when necessary. By unifying training, deployment, monitoring, and physical testing into a single framework, the research intends to provide a scalable, reliable, and adaptable solution that can serve as a benchmark for future developments in autonomous navigation for both industrial and service robotics contexts.

2.2 Specific Objectives

2.2.1 Development of a Model Training Platform

To build a comprehensive training platform capable of supporting the iterative development of navigation models. This platform will simulate varied indoor environments with dynamic obstacles, allowing for extensive testing of SLAM, path planning, and obstacle avoidance algorithms. It will support reinforcement learning and other adaptive approaches by providing a controlled yet highly variable training environment. The platform will incorporate automated preprocessing pipelines, dataset generation tools, and performance benchmarking modules. By enabling hyperparameter tuning, distributed training, and modular integration of different algorithmic approaches, the platform will foster rapid experimentation and refinement. Its output will be standardized, ensuring that trained models can seamlessly move into the deployment phase without requiring extensive reconfiguration. The training process will be designed to capture edge cases and environmental anomalies, ensuring that models generalize well to real-world operational conditions.

2.2.2 Development of the Deployment Platform

This focuses on establishing a reliable deployment platform that bridges the gap between trained models and operational use. This will involve designing software infrastructure to manage the transfer of models from the training environment to the field without compatibility issues. The platform will handle version control, rollback mechanisms, and over-the-air updates to ensure rapid and safe deployment. It will also provide integration interfaces for hardware, enabling compatibility with a wide range of sensor configurations and computational units. The deployment pipeline will prioritize operational readiness, meaning deployed models must meet predefined benchmarks before activation. This approach ensures minimal downtime and reduces the risks associated with deploying untested updates. The deployment platform will also be designed for scalability, allowing multiple robots or units to be updated and managed simultaneously.

2.2.3 Real-time Monitoring and Performance Evaluation

Goal is to create a robust monitoring and evaluation framework that tracks the operational performance of deployed models in real time. This will involve continuous collection of sensor data, navigation metrics, and system health parameters, which will be visualized through an operator dashboard. The system will feature automated anomaly detection to identify deviations from expected performance, triggering alerts or adaptive responses. Performance trends will be logged over time to enable predictive maintenance and gradual model refinement. This monitoring framework will act as a feedback loop to the training platform, ensuring that the system evolves based on real-world operational data. By enabling proactive issue resolution and continuous improvement, the monitoring system will ensure that the autonomous navigation platform remains reliable, efficient, and adaptable to changing environments.

2.2.1 Design and Implement a Modular SLAM Backend

Design and implement a scalable, server-side backend capable of processing and managing data from a fleet of robots. This backend will feature a modular, pluggable architecture with well-defined APIs to ingest sensor data from multiple sources. It will be responsible for fusing partial maps from individual robots into a consistent global 3D map, storing historical data for later analysis, and implementing the core change detection algorithms to compare map versions and identify environmental modifications.

2.2.2 Develop an Interactive 3D Visualization Client

Create a high-performance, web-based interface for visualizing the multi-robot system in real time. Using modern 3D rendering libraries, this client will display the fused SLAM-generated maps and live telemetry from all robots. The focus will be on creating an intuitive and responsive user experience that allows

operators to easily navigate complex 3D scenes and understand the state of the system at a glance.

2.2.3 Engineer a Low-Latency Data Synchronization Gateway

Goal is to engineer an efficient, low-latency communication gateway to stream data from the backend to the visualization client. This real-time link is critical for providing operators with up-to-the-second information on robot positions, alerts about detected map changes, and other system notifications. This will involve implementing modern communication protocols like WebSockets and optimizing data serialization to minimize bandwidth and ensure a smooth user experience.

2.2.4 Integrate Advanced Debugging and Analytics Tools

Objective is to move beyond simple monitoring by integrating advanced diagnostic and analytical tools into the platform. This includes developing features for visualizing historical data, such as collision heatmaps, and providing a version-controlled "map diff" tool to visually compare different states of the environment. This will allow operators to see exactly what has changed over time, such as new obstacles or altered pathways, which is critical for debugging and maintaining trust in the system's perception.

2.2.5 Build a User-Interactive Annotation System

Empower operators with the ability to interact directly with the environmental map. This involves building tools within the interface that allow users to tag known obstacles, define operational zones (e.g., "keep-out" areas), or manually confirm or reject changes detected by the backend. This feature creates a human-in-the-loop feedback mechanism, enhancing the semantic understanding of the environment for both the robots and human operators.

2.2.6 Development of a Robust Communication Infrastructure

Communication system that allows robots to share partial maps and sensor data in real-time. This includes:

- Implementation of ROS 2 topic-based communication for efficient sensor data and map broadcasting
- Namespace-based isolation of each robot's data to prevent conflicts and ensure scalability
- Efficient topic subscription and processing to minimize bandwidth usage and latency

This communication layer has been designed to support teams of multiple robots simultaneously, allowing efficient coordination without sacrificing performance.

2.2.7 Development of an Accurate and Scalable Map Fusion System

Create an incremental map fusion process that can merge local maps created by separate robots into a consistent, shared global map. This requires:

- Implementation of ICP (Iterative Closest Point) algorithms for accurate map alignment.
- TF (Transform Framework) integration for proper coordinate frame management between robots.
- Efficient update mechanisms that integrate new mapping data into the global map without interrupting ongoing operations.
- Support for variable map resolutions and grid sizes to accommodate different environmental scales.

The resulting global map is continuously generated and updated, ensuring that all robots have access to the most current environmental data.

2.2.8 Dynamic Task Coordination and Collision Avoidance

Implement mechanisms for detecting and preventing navigation conflicts between multiple robots. This includes:

- Obstacle detection using sensor data from all robots in the collaborative environment.
- Shared costmap representation that identifies navigable and blocked areas.
- Collision detection algorithms that identify potential path conflicts before they occur.

This dynamic coordination minimizes collision risks, maximizes coverage efficiency, and maintains operational fluidity in unpredictable environments.

2.2.9 Integration with Real-Time Visualization Platform

Establish seamless data transmission from the multi-robot simulation framework to an external SLAM Visualization Platform. This integration provides:

- Real-time transmission of robot positions, map updates, and exploration status to the visualization platform.
- Web-based interfaces displaying merged maps, robot trajectories, and collaborative exploration metrics.
- Operator tools for monitoring team performance, detecting hazards through heatmap analysis, and annotating areas of interest.
- Bidirectional communication enabling operator feedback to influence robot task assignments and priorities.

These integrations enable comprehensive operational oversight while providing a research platform for analyzing multi-robot collaboration effectiveness.

2.2.10 Develop a Configurable Virtual Environment for Data Generation

- Using a target set of typical home or warehouse items, create one or more intricate indoor scenes in the Gazebo simulator.
- To build a model of a camera-equipped robot in URDF (Unified Robot Description Format) and include it into the virtual world.

2.2.11 Implement Automated Synthetic Data Generation Pipeline

- To create scripts that gather visual data from the robot's virtual camera automatically.
- To use procedurally changing textures, lighting, and item placements in the simulation to use domain randomization approaches.
- To guarantee that all target objects in the collected photos have pixel-perfect bounding box annotations automatically generated by the pipeline.

2.2.12 Train and Optimize a State-of-the-Art Object Detection Model

- To train a YOLOv8 object identification model using the artificial dataset that was created.
- To maximize the accuracy and generalization potential of the model by experimenting with various training hyperparameters and data augmentation techniques.

2.2.13 Deploy and Validate the Trained Model in a ROS 2 Environment

- To create a Python node for ROS 2 that loads the custom model that has been trained.
- In addition to doing real-time object identification and publishing the findings (class labels, confidence scores, and bounding boxes) to a specific ROS 2 topic, this node will subscribe to a simulated camera feed.
- To verify the model's performance both qualitatively through RViz2 visualization and statistically using common measures (mAP, IoU).

3 METHODOLOGY

Implementing an autonomous navigation platform that includes SLAM, Meta-RL, and multi-robot coordination. The system facilitates decentralized collaboration among robots, allowing for real-time mapping data exchange, synchronized navigation decisions, and conflict resolution in dynamic indoor situations. The workflow follows a staged approach that includes requirement analysis, system design, implementation, testing, and optimization. Each phase addresses issues such as dynamic environmental changes, network dependability, computational restrictions and scalability, ensuring that the framework is adaptable and efficient in real-world applications.

3.1 System Overview

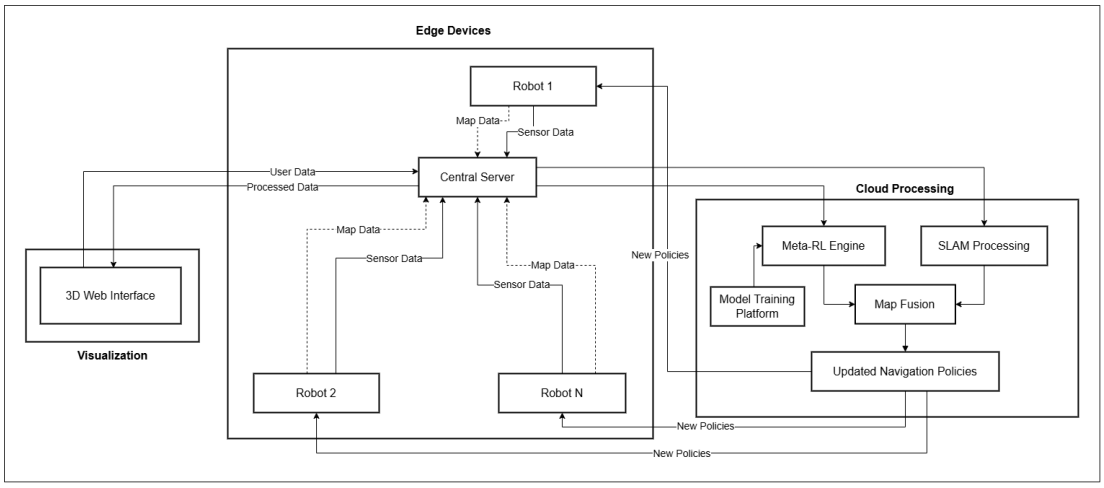

Figure 3.1 System Architecture Diagram

Edge Devices: The framework consists of several robots, each equipped with cameras, LiDAR, and other sensors (e.g. IMUs) for collecting raw maps and sensor data. These edge devices have minimal onboard processing capabilities and are intended primarily to capture and transmit data to the central server rather than perform complex calculations. To ensure high-quality input, robots use robust data gathering techniques, while simple connectivity protocols allow real-time data streaming to the server.

Central Server: The central server functions as the main processing unit, receiving raw map and sensor data from all robots. It performs heavy lifting tasks such as real-

time map creation and fusion, as well as dynamic task allocation using collaborative SLAM techniques. The server collects and interprets incoming data to generate a consistent global map, resolves navigation conflicts, and assigns exploration zones based on robot telemetry (for example, battery condition). User data from the visualization interface is also analyzed here to aid with manual interventions such as work reassignment.

Cloud Processing: The cloud layer provides the system with powerful computing resources, including the Meta-RL Engine, SLAM Processing, Map Fusion, and Modeling Platform modules. The Meta-RL Engine trains adaptive navigation policies and aggregated data from the central server. SLAM Processing uses large-scale optimization approaches to update global maps whereas the Map Fusion module combines local data into a consistent representation that accounts for dynamic changes. The Modeling Platform simulates scenarios to develop new rules, which are then sent to the robots as updated navigation policies using the central server.

Visualization: The 3D Web Interface is a real-time visualization and control platform that receives processed data from a central server. It displays robot movements, global map updates, and job assignments, and includes features such as collision heatmaps. Operators can leverage user data such as priority zone marks to affect job assignments, allowing for greater flexibility in mission management.

This architecture ensures effective data handling and interoperability, with the central server and cloud processing reacting to network fluctuations and changes in the environment. It uses robust SLAM approaches and cooperative navigation strategies, making it appropriate for applications such as warehouse automation and smart houses.

Simulation Environment (Gazebo): This is the foundation of the pipeline. A 3D virtual world is created containing the indoor layout, furniture, and a set of target objects to be identified. A simulated mobile robot, equipped with a virtual camera, is placed within this environment.

Data Generation & Augmentation Engine: A collection of Python scripts that manages the simulation and produces a wide range of data.

- Scenario Controller: To take pictures from multiple angles, this script programmatically moves the robot to different places and orientations.
- Domain Randomizer: This module modifies the simulation's attributes at each stage. It might alter the lighting's hue and intensity, switch up the wall and floor textures, or shift the things a little.
- Data Collector & Annotator: This script stores the camera's pictures and, most importantly, asks the Gazebo simulation engine for the exact 2D bounding boxes of the objects that are in the camera's field of vision. The YOLO annotation format is used to store this data.

Robot SLAM Nodes: Each robot in the system runs an independent SLAM instance using SLAM Toolbox operating in asynchronous mode. This configuration enables continuous, non-blocking map updates that are decoupled from the robot's real-time navigation cycle, ensuring that map-building operations do not introduce latency into locomotion or sensor processing pipelines.

Each robot is assigned to its own ROS 2 namespace for example, /robot1, /robot2, through to /robotN which serves to fully isolate its topics, services, and TF frames from those of other agents in the fleet. This namespace-based isolation is critical in multi-robot deployments, as it prevents topic collisions and TF tree conflicts that would otherwise corrupt localization estimates across the system.

Robots are equipped with onboard LiDAR sensors whose scan data is continuously published to their respective /robotN/scan topics. Odometry information, derived from wheel encoders or IMU fusion, is consumed by each robot's SLAM instance to track displacement and correct for accumulated drift. From these inputs, SLAM Toolbox constructs and incrementally refines an individual occupancy grid map for each agent, published under the /robotN/map topic. Each occupancy grid encodes a probabilistic representation of the surrounding environment, classifying cells as free, occupied, or unknown based on accumulated sensor evidence. These per-robot maps serve as the primary inputs to the downstream Map Fusion Engine, which aggregates them into a coherent global representation shared across the entire fleet.

Map Fusion Engine: A centralized map merging node sits at the core of the multi-robot collaboration architecture, continuously receiving individual occupancy grid maps from all active robots and fusing them into a single, unified global map. This component is responsible for resolving the geometric and probabilistic inconsistencies that naturally arise when multiple agents independently observe overlapping regions of the same environment from different vantage points.

The fusion process begins with coordinate frame alignment. Each robot's local map is expressed in its own origin frame, so TF transform lookups are used to project individual maps into a shared common coordinate frame, typically the global /map frame before any merging takes place. This step ensures spatial consistency and prevents misaligned overlays that would introduce ghost obstacles or erase valid free-space readings.

Once maps are aligned into a common reference, the Iterative Closest Point (ICP) algorithm is applied to perform fine-grained map registration. ICP iteratively minimizes the spatial error between overlapping point distributions from different robot maps, correcting for residual misalignments caused by odometry drift or minor localization discrepancies. This registration step is particularly important in environments with repeated structural features, such as warehouse aisles, where coarse alignment alone is insufficient to produce geometrically accurate fused maps.

Following registration, a probabilistic cell merging strategy reconciles conflicting occupancy values for grid cells observed by multiple robots. Cells are merged using a log-odds update scheme, which weights observations according to sensor confidence and recency, producing a global occupancy grid that reflects the best collective understanding of the environment at any given moment. The merged map is maintained and updated in real-time as new robot map data arrives, ensuring that the global representation remains current and accurate throughout the exploration mission.

Gazebo Simulation Environment: The full multi-robot framework is validated within the Gazebo simulation environment, which provides a physics-based, high-fidelity virtual testbed for the entire system prior to real-world deployment. Gazebo replicates the dynamics of physical robot hardware including LiDAR scan patterns,

odometry noise characteristics, and collision responses, allowing the framework to be exercised under conditions that closely approximate those encountered in operational settings.

For this project, Gazebo is configured with a realistic warehouse environment that includes static structural elements such as shelving units, perimeter walls, loading bays, and floor-level obstacles. This layout presents the multi-robot system with a representative exploration challenge: a large, partially occluded space with narrow corridors and open areas that demand coordinated coverage strategies to map efficiently. The static nature of the simulated obstacles provides a controlled baseline against which map fusion accuracy, task allocation efficiency, and conflict resolution performance can be measured without the confounding variability introduced by dynamic elements.

Within this environment, multiple robot models are each instantiated with its own namespace, SLAM instance and sensor suite that are spawned and coordinated through the same software stack that will govern physical robot deployments. This design ensures that simulation results are directly transferable to real-world trials with minimal reconfiguration. The simulation environment additionally supports rapid iteration over robot fleet configurations, exploration strategies and communication parameters, enabling systematic evaluation of system performance across a wide range of experimental conditions without the time and cost overhead of physical hardware testing.

Visualization: A ROS 2 web bridge component acts as the communication link between the backend map fusion engine and the SLAM Visualization Platform, enabling real-time situational awareness for human operators monitoring the fleet. The web bridge serializes ROS 2 topic data including merged map updates, robot pose estimates, sensor readings and system health metrics into a format suitable for transmission over WebSocket connections to browser-based dashboard clients.

The visualization platform receives and renders a continuous stream of merged global map updates, providing operators with a live, evolving picture of the environment as it is explored and mapped by the robot fleet. Robot positions are overlaid directly onto

the global map using pose data drawn from each agent's localization output, allowing operators to track individual robot trajectories and assess spatial coverage in real time.

In addition to map and position data, the platform displays collision detection outputs, highlighting areas where robot paths come into proximity or where the conflict resolution module has intervened to re-route agents. Heatmap overlays are generated from historical robot trajectory and sensor data, visually encoding metrics such as visitation frequency, mapping confidence, and communication latency across different regions of the environment. These heatmaps serve as diagnostic tools, enabling operators to identify under-explored zones, detect recurring navigation bottlenecks, and assess the overall efficiency of the fleet's exploration strategy. Together, these visualization capabilities provide the operational transparency required for effective human oversight of a complex, autonomous multi-robot system.

3.2 Software Solutions

3.2.1 Requirement Gathering and Analysis

The development process begins with gathering precise requirements for the model training platform, deployment platform, and prototype robot, ensuring that trained navigation models can transition seamlessly from simulated environments to real-world execution. Key requirements include determining the computational resources necessary for real-time processing, defining the complexity of simulated environments, and designing a robust communication pipeline between the training system, deployment hardware, and onboard sensors.

During this stage, available LiDAR datasets such as open-source urban mapping and indoor scan repositories are evaluated to quickly test algorithms and assess processing pipelines without immediate reliance on physical hardware. This early experimentation helps identify optimal preprocessing techniques and filtering methods

to handle issues such as sensor noise, occlusions, and varying lighting conditions for camera data.

To support informed sensor and algorithm selection, a prototype 3D visualization tool will be developed. This tool will render LiDAR point clouds and camera frames together, enabling real-time inspection of spatial accuracy, alignment quality, and environmental coverage. Such visualization aids in determining the most suitable LiDAR resolution, field of view, and camera lens configuration for the final robot.

Requirement gathering also draws on industry standards, academic literature, and small-scale proof-of-concept trials, ensuring that each component aligns with proven best practices. Potential failure modes such as inaccurate SLAM mapping due to reflective surfaces or delays in model updates due to network latency are identified early to inform mitigation strategies. By finalizing these requirements upfront, the subsequent design and development stages can progress with reduced risk, streamlined integration, and greater confidence in achieving the project's navigation goals.

3.2.2 Feasibility Study

Schedule Feasibility

The proposed project schedule is designed to be both realistic and efficient, following a phased and partially overlapping approach to reduce overall development time. Phase one will focus on the design and implementation of the simulation and model training platform, where synthetic datasets will be generated using simulated LiDAR and camera feeds within Gazebo. This will allow early-stage algorithm development without waiting for hardware readiness. Phase two will begin while phase one is still ongoing, dedicated to the creation of the deployment platform and the initial assembly of the prototype robot. This overlap ensures that software readiness coincides with

hardware availability. The third phase will integrate all components and model training outputs, deployment infrastructure, and robot hardware followed by iterative testing in real-world environments. These tests will include varied indoor and semi-outdoor conditions to evaluate model adaptability. By planning for parallel task execution, leveraging existing open-source tools, and setting milestone-based progress reviews, the project maintains a balance between speed and quality assurance. Contingency periods are built into the timeline to address unforeseen hardware delays, software debugging, or integration complexities. The schedule is further strengthened by the use of containerized deployment pipelines, which significantly reduce integration time between the training environment and operational platform.

Technical Feasibility

The technical feasibility of the project is high, as it leverages well-established robotics frameworks and hardware platforms. The prototype robot will be based on an Arduino microcontroller for sensor integration and control, providing a lightweight and cost-effective hardware solution. The model training and inference will be hosted on a cloud platform, enabling powerful computation without relying on onboard GPUs or TPUs. Sensor data, including inputs from a 360° LiDAR sensor and a monocular or stereo camera, will be transmitted from the Arduino to the cloud over a reliable network connection. There, deep learning models will process the data to perform tasks such as SLAM, object detection, and navigation planning. The cloud-based approach allows the use of computationally demanding algorithms without being constrained by the robot's onboard hardware. The modular design of the system ensures that individual components, such as sensors or communication modules, can be swapped or upgraded without major architectural changes. Standardized sensor message formats and established robotics protocols ensure smooth integration between the Arduino and the cloud services. Additionally, leveraging off-the-shelf sensors and Arduino hardware reduces procurement risks, while the use of open-source cloud tools minimizes software licensing costs. The compatibility of frontend web technologies (e.g., WebGL, Three.js) with target browsers for rendering complex 3D scenes. Evaluating

backend frameworks (e.g., Python/Flask, Node.js/Express) for their ability to process large datasets and stream data efficiently via WebSockets.

As a potential upgrade, the Arduino-based robot can be enhanced by integrating an NVIDIA Jetson Orin Nano or similar embedded AI board. This would enable local processing of LiDAR point clouds and camera imagery, reducing reliance on cloud computation and network latency. With sufficient GPU compute power, the onboard board could execute SLAM, object detection, and navigation algorithms in real time, opening the door for more autonomous operation and faster response times. This step would improve system performance while retaining the modular and upgradeable architecture. All the core software components (Ubuntu, ROS 2, Gazebo, PyTorch, YOLOv8) are mature, open-source, and well-documented, with proven compatibility. The primary technical challenge training a deep learning model is well within the capabilities of modern consumer-grade GPUs. The process of integrating a trained model into a ROS 2 node is a standard and well-understood task in the robotics community.

Economic Feasibility

Estimating costs for hardware procurement (robot units, sensors, server hardware), cloud computing resources, cloud infrastructure (server hosting, database storage), web hosting, and potential software licensing for specialized 3D libraries or UI components, and software licensing if applicable.

Operational Feasibility

Assessing integration complexity with existing SLAM and Meta-RL systems, operator training requirements, and maintenance demands. The final output self-contained ROS 2 node is designed for modular integration with the larger team project. The skills required to complete the project (Python programming, Linux command line, basic

deep learning concepts) are aligned with the curriculum of a Software Engineering specialization. Extensive documentation and tutorials for all chosen technologies will mitigate the learning curve.

3.3 Tasks and Sub-Tasks

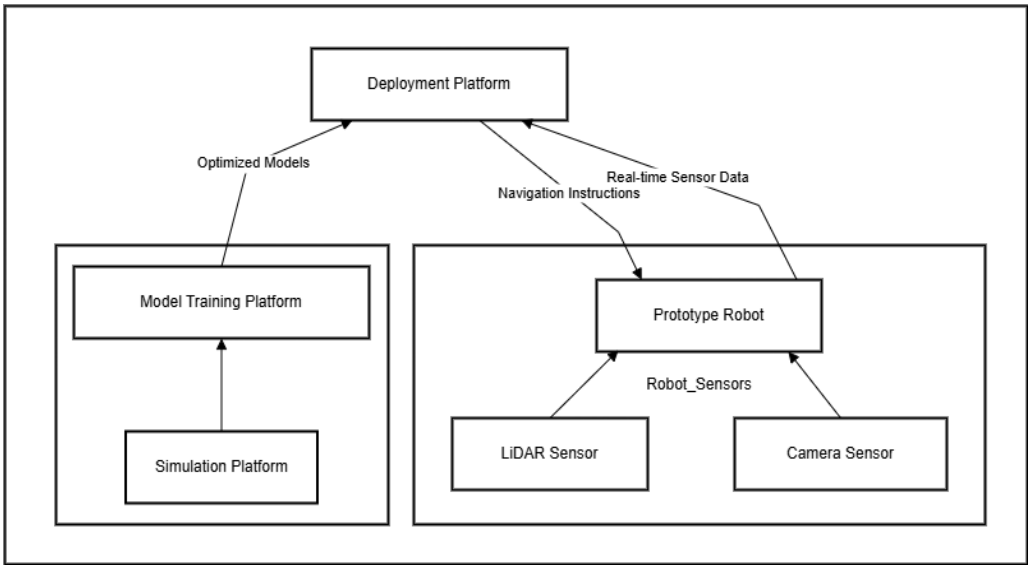

Figure 3.2 Meta RL model deployment platform Diagram

In order to accomplish the Multi-Robot Collaboration Framework project goals, the development process is divided into large work packages, each of which is further subdivided into distinct tasks to guarantee steady advancement and effective execution. These activities are organized to meet technical, operational, and validation criteria and are intended to be in line with the project timeframe. The system consists of three interconnected modules. The model training platform runs in a controlled simulation environment where LiDAR and camera inputs are emulated to mimic real-world conditions. The deployment platform acts as a bridge, transferring optimized models to the robot and enabling live monitoring. The prototype robot integrates the LiDAR and camera sensors, feeding real-time data into the onboard navigation stack. Data pipelines ensure that sensor data in simulation matches real-world formats, enabling smooth model transfer.

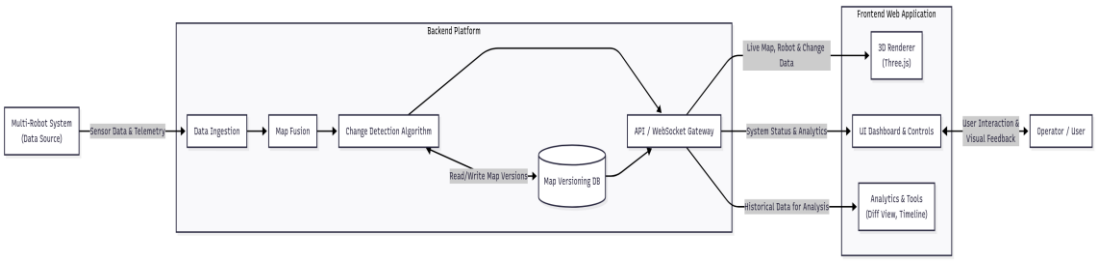

Figure 3.3 Component Architecture of the Visualization Platform

This component, the 3D Mapping & Visualization Platform, serves as the central hub for environmental state management and human operational oversight. It is a full-stack application with two primary parts: a backend for data processing and a frontend for visualization.

The backend is responsible for receiving sensor data from the entire robot fleet via the central server. Its first task is to fuse the partial maps from each robot into a single, consistent global map. The core logic then resides in its change detection pipeline. This pipeline continuously compares new incoming data against the current version of the global map to identify structural changes. If a change is validated, the backend creates and stores a new, time-stamped version of the map, effectively maintaining a historical record of the environment.

The frontend is an interactive, web-based 3D client. It communicates with the backend via a low-latency gateway to receive and render the live global map and the real-time positions of all robots. Its most critical feature is the visualization of the data produced by the change detection backend. Operators can use tools like a "map diff" view to see highlighted changes between versions or use a timeline to review the environment's evolution. This tight integration of backend detection and frontend visualization makes the system a powerful tool for monitoring and diagnostics.

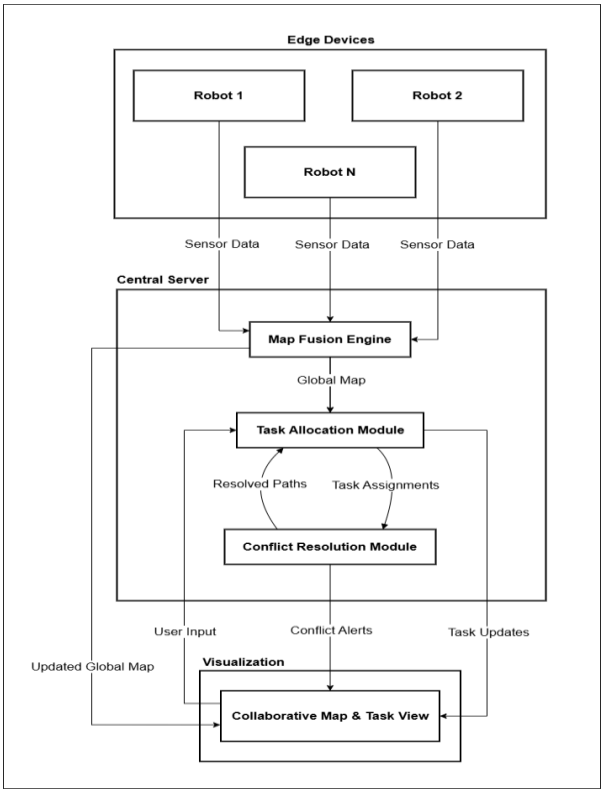

Figure 3.5 Multi Robot Collaboration Framework

In addition to the complete system overview shown in Figure 3.1, the Multi-Robot Collaboration Framework's internal design can be represented as a distinct component structure as seen in Figure 3.4. While the overall system integrates Gazebo simulation and visualization layers, this sub-architecture highlights the specific modules that enable real-time cooperation between multiple robots in collaborative mapping and exploration tasks.

The primary function of the robot edge devices is to gather raw sensor data from LiDAR sensors and transmit this data through ROS 2 topics to the local SLAM processing modules. Each robot runs its own SLAM Toolbox instance, which processes the continuous stream of sensor observations asynchronously. By operating in asynchronous mode, individual SLAM instances avoid blocking each other, allowing the system to scale efficiently with additional robots. The SLAM Toolbox on each robot generates local occupancy grid maps representing that robot's

understanding of the environment, published at 1 Hz to balance accuracy with network efficiency.

These local maps are then forwarded to the central Map Fusion Engine, which serves as the coordination hub of the framework. The Map Fusion Engine minimizes inconsistencies brought on by odometry drift or sensor noise by combining incoming local maps into a single, coherent global map. This process uses established techniques such as Iterative Closest Point for accurate map registration and Transform Framework transforms for coordinate frame alignment, techniques which have proven effective in multi-robot mapping scenarios [24]. The fusion engine continuously receives new map observations from all active robots through ROS 2 topic subscriptions and incrementally updates the global map, ensuring that all robots have access to the latest collective understanding of the environment without interrupting ongoing exploration activities. The incremental approach maintains efficiency and each new robot map is registered to the existing global map rather than re-registering all maps from scratch.

Alongside real-time map fusion, the Collision Detection and Heatmap Generation Module analyze the merged global map to identify obstacles, walls, and potential navigation conflicts. This module examines occupancy grid cells to detect high-density obstacle areas and generates heatmap visualizations showing regions where robots frequently interact or where collisions are more likely to occur. The collision detection process operates continuously on the merged map, providing real-time awareness of hazardous zones and navigation bottlenecks. When potential conflicts are detected, the system flags these areas to operators and provides visualization feedback that can guide robot exploration patterns to avoid congestion. This capability is essential for maintaining safe multi-robot operation in constrained environments.

The Web Bridge and Visualization Integration Module receive all outputs from the framework modules and transmits this rich data to an external SLAM Visualization Platform in real-time. The web bridge implements JSON serialization for maps, robot positions, and collision metrics, transmitting this data via HTTP REST endpoints and WebSocket connections for real-time updates. To maintain a balance between full automation and human oversight, the visualization interface enables operators to step

in as necessary such as annotating hazardous regions, marking priority exploration zones, or adjusting robot mission parameters based on real-time observations. This human-in-the-loop capability is essential for adapting to unexpected environmental conditions or operational priorities that emerge during exploration. Operators can examine robot positions, merged maps at multiple zoom levels, collision heatmaps highlighting risk areas, and exploration progress metrics with comprehensive interactive controls.

The ROS 2-based communication layer underlying these modules ensures reliable, scalable data exchange between robots and the central fusion infrastructure. Topic-based pub-sub messaging provides loose coupling between components, allowing independent development and testing of individual modules while maintaining clear data flow pathways. Namespace-based isolation prevents conflicts when multiple robots are active simultaneously, and TF broadcasting ensures consistent coordinate frame references across all robots and the global map. This architecture enables the system to gracefully handle 2-5 robot teams while maintaining acceptable computational overhead

This sub-architecture demonstrates how the Multi-Robot Collaboration Framework functions as the system's collaboration center. It enables robots to collaborate effectively by sharing observations, merging maps into a unified representation, and coordinating their exploration strategies to maximize coverage while avoiding conflicts. By combining real-time map fusion with collision detection, visualization tools, and operator interaction capabilities, the framework facilitates practical applications in search-and-rescue operations, warehouse automation, facility inspection, and other multi-robot coordination scenarios. The framework's emphasis on real-time data synchronization and visualization creates an operational platform where robot teams and human supervisors work together as an integrated system to accomplish complex exploration tasks efficiently and safely.

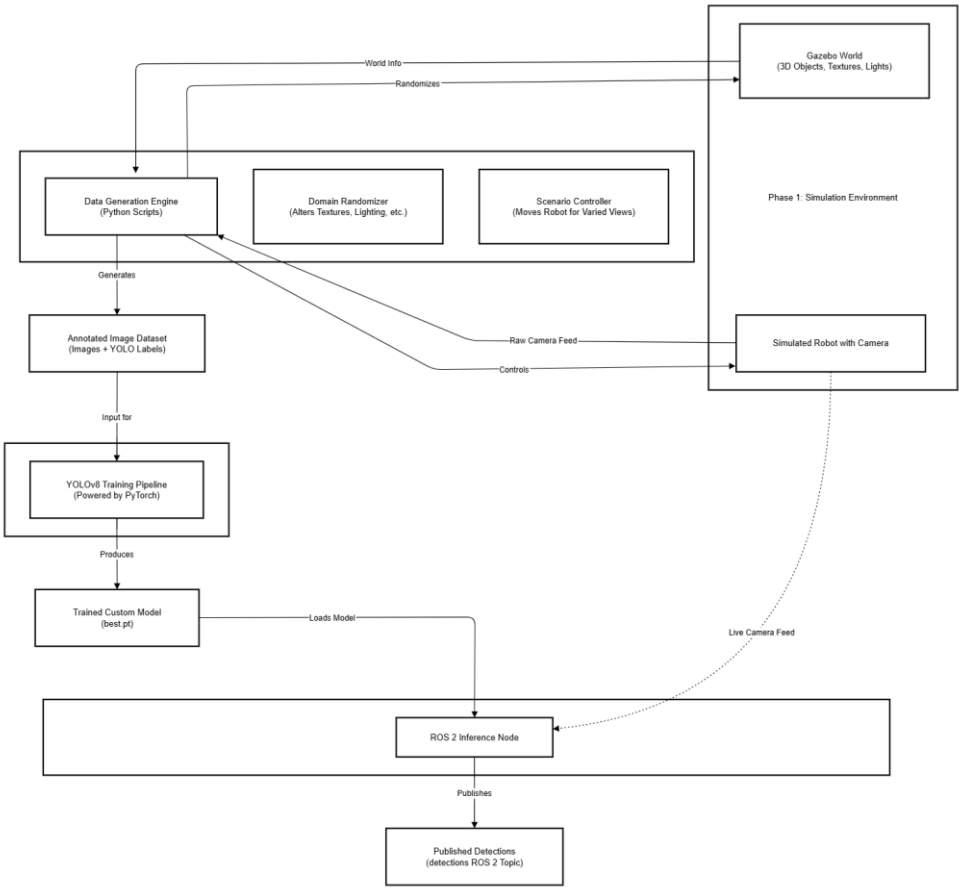

Figure 3.6 Simulation platform

Several common, open-source tools that are extensively utilized in the robotics and artificial intelligence sectors serve as the foundation for this component. This guarantees stability, compatibility, and availability of a large community of support. The planning process starts with a detailed requirements analysis, which is followed by a study to make sure the project is doable given its technical and schedule limitations.

1. **System Architecture & Prototype Setup**
   - Assemble and Configure Prototype Robots: First prototype comes together by combining a camera, a processing unit, and an Arduino-based control system

into a simple but functional loop. The camera acts as the primary sensor, continuously capturing visual input from the environment. These frames are processed on a laptop or Raspberry Pi, where the meta-reinforcement learning policy interprets the scene and decides the appropriate action, such as moving forward, turning, or stopping. The decision is then transmitted to the Arduino via wi-fi, which serves as the executor by converting high-level commands into low-level motor driver signals that control the wheels or tracks of the robot. To ensure safety during early tests, an LIDAR sensor can be connected directly to the Arduino, providing an emergency stop if obstacles are detected at close range. This modular setup vision and intelligence on the laptop or Pi, execution on the Arduino, and mobility through motors allows for a practical and scalable foundation, enabling quick iteration while leaving room for wireless communication and advanced sensors in later versions.

2. **Software Module Development**
   - Model training platform implementation.
   - Communication layer design.
   - Model deployment implementation.
3. **Integration & Deployment**
   - Integrate central server processing with cloud optimization.
   - Integrate simulation platform with training platform .
4. **Testing & Validation**
   - Unit and integration tests.
   - Scalability testing with multiple robots.
   - Real-world trials in controlled environments.
5. **Documentation & Handover**
   - System architecture documentation.
   - User manual creation.

3.4 Testing and Validation

Testing will be performed in progressive stages:

- **Unit Testing:** Each software module (map fusion, task allocation, conflict resolution) will be tested independently to verify functionality and accuracy.
- **Integration Testing:** Data flow between robots, central server, cloud, and visualization interface will be validated to ensure seamless interaction.
- **Performance Testing:** Metrics such as map fusion latency, task allocation fairness, and communication bandwidth usage will be measured to assess efficiency.
- **Scalability Testing:** Simulations with larger robot fleets under varied network and environmental conditions will evaluate system robustness.
- **Fault Tolerance Testing:** Network links will be disconnected to confirm autonomous fallback to local SLAM, ensuring safety during disruptions.

This staged approach ensures comprehensive validation across all components.

3.5 Data Collection

Data will be gathered from both simulated and real-world environments to support the development and validation of the Multi-Robot Collaboration Framework.

- **Simulated Data:** Generated using simulations to create diverse indoor scenarios for algorithm training, providing a controlled environment to test SLAM and Meta-RL models with varying conditions.
- **Real-World Data:** Collected from robot sensors in controlled indoor settings, such as university labs or mock warehouse spaces, to capture authentic environmental data and sensor noise.

The collected data will be processed into synchronized datasets, aligning sensor inputs with timestamps for use in SLAM map fusion and policy training, ensuring compatibility with the central server and cloud infrastructure.

3.6 Time Frame & Schedule

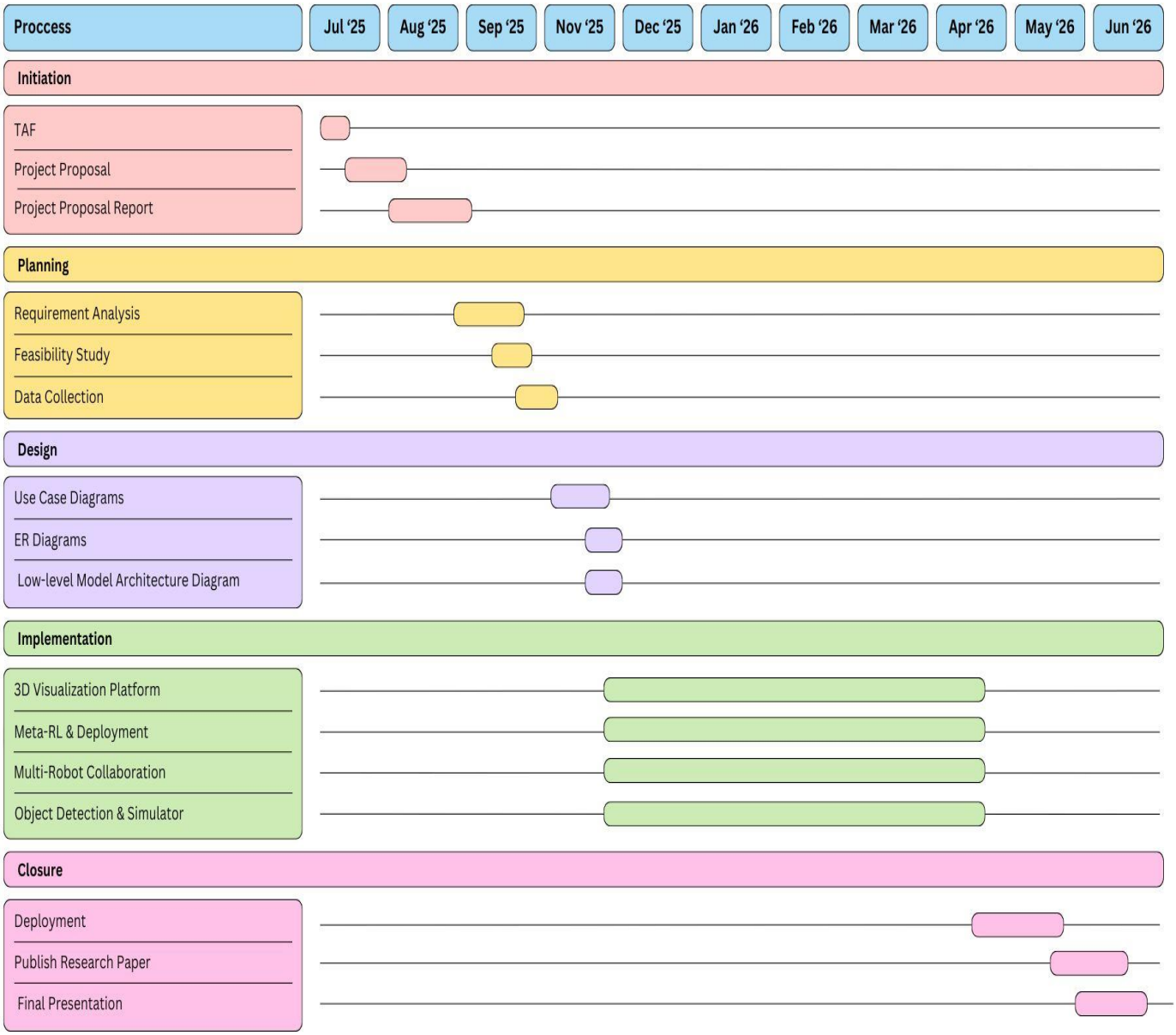

Figure 3.4 Gantt Chart

4 PROJECT REQUIREMENTS

4.1.1 Functional Requirements

The functional requirements define the specific features and capabilities that the system must deliver to achieve seamless model training, deployment, and real-world navigation. For the model training platform, the system must support importing LiDAR and camera data from both simulated and real environments, preprocessing it into synchronized datasets, and training deep learning or reinforcement learning models for path planning and obstacle avoidance. The platform should allow iterative training, where new data from field tests can be fed back into the training pipeline to improve accuracy.

For the deployment platform, the system must enable smooth transfer of trained models to the onboard computing unit of the robot, with minimal conversion or compatibility issues. It must support real-time inference from LiDAR and camera streams, integrating sensor fusion techniques to maintain consistent spatial awareness.

For the prototype robot, the system must integrate the LiDAR and camera sensors physically and logically, ensuring accurate calibration, time synchronization, and robust data transmission to the onboard processor. The robot must execute navigation commands in real time based on model outputs, and be capable of autonomous obstacle detection, mapping, and rerouting. Additionally, a real-time monitoring interface is required to visualize live sensor data, current navigation paths, and system diagnostics.

The system must be able to automatically generate a dataset of images and corresponding bounding box annotations from the Gazebo simulation. The system must use the generated dataset to train a custom YOLOv8 model capable of detecting a predefined set of objects. The deployed ROS 2 node must be able to process a live camera feed and detect objects in real-time. The node must publish the detection results including class name, confidence score, and bounding box to a ROS 2 topic. The detection results must be visualizable in RViz2, showing bounding boxes overlaid on the camera image.

These functional requirements ensure that the system not only works within the lab environment but also transitions effectively to dynamic real-world conditions, supporting the iterative cycle of simulation, deployment, testing, and model refinement.

4.1.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance constraints that the system must meet to ensure reliability, scalability, and usability. For the model training platform, performance and efficiency are critical. Training must be optimized to run within acceptable time frames, even with large LiDAR datasets, and the system must support parallel processing on GPUs or cloud-based compute clusters. The platform should also have modularity, allowing easy integration of new algorithms, sensors, or preprocessing techniques without major reengineering.

For the deployment platform, latency must be minimal to ensure that LiDAR and camera data are processed fast enough for real-time navigation. The system should be fault-tolerant, handling temporary sensor dropouts or degraded network conditions without significant performance loss. Security is another key aspect model files and deployment pipelines must be protected from unauthorized access or tampering.

For the prototype robot, robustness and durability are essential. The hardware must be capable of operating in various environments, from indoor spaces with tight corridors to outdoor areas with uneven terrain. Power efficiency is important to extend operational runtime, and thermal management must ensure stable processing under continuous load.

Usability is also a priority. The training and deployment platforms should have clear, intuitive interfaces for monitoring performance, adjusting parameters, and triggering retraining. Documentation and maintainability requirements ensure that future developers can understand, modify, and extend the system without steep learning curves. Collectively, these non-functional requirements guarantee that the system performs consistently, scales effectively, and remains adaptable for future research and real-world deployment scenarios.

The inference node should achieve a processing speed of at least 10 frames per second (FPS) on the target hardware to be considered "real-time." The trained model should achieve a mean Average Precision (mAP@.50) of at least 0.75 on the synthetic test

dataset. The data generation scripts, training code, and ROS 2 node should be well-structured and independent, allowing them to be easily modified or reused. The entire pipeline should be well-documented, allowing another researcher to replicate the data generation and training process.

4.1.3 User Requirements

Operators must use an easy-to-use interface to oversee and manage missions: Clear controls and graphics should be provided using the interface. Ability to override automatic decisions in an emergency: To improve safety and flexibility in critical situations, operators must have the ability to manually modify tasks or pathways.

4.1.4 System Requirements

Robots with LiDAR, RGB-D cameras, and IMUs: In order to gather unprocessed data and send it to the central server, edge devices need to be outfitted with these sensors. Central server with GPU acceleration: To effectively manage demanding map fusion and decision-making activities, the server needs GPU assistance. Cloud infrastructure for policy training: Meta-RL training and large-scale SLAM optimization with aggregated data require a cloud system. For the simulation platform computer running Ubuntu 22.04 with ROS 2 Humble installed. A CUDA-enabled NVIDIA GPU is required for efficient training of the YOLOv8 model. Python 3.8+, PyTorch, and all libraries specified in the Ultralytics YOLOv8 requirements.

5 RESULT & DISCUSSION

5.1 Results

The SLAM backbone and map fusion engine demonstrated consistent performance across multi-robot team configurations ranging from two to five robots operating simultaneously within the Gazebo warehouse simulation environment. Individual robots, each running an independent SLAM Toolbox instance in asynchronous mode, successfully generated local occupancy grid maps published at 1 Hz without introducing blocking latency into their respective navigation cycles. The Map Fusion Engine, receiving these per-robot maps via ROS 2 topic subscriptions, applied TF-based coordinate frame alignment followed by ICP registration to resolve inter-robot spatial inconsistencies arising from odometry drift and differing starting orientations. The resulting global occupancy grid maintained high spatial coherence across all active robot namespaces, with the log-odds probabilistic cell merging strategy effectively reconciling conflicting sensor readings in regions observed simultaneously by multiple agents. Multi-robot teams achieved substantially higher area coverage within equivalent time windows compared to single-agent baselines, confirming that decentralized collaborative exploration reduces redundant traversal and accelerates environmental mapping in partially occluded spaces such as narrow warehouse corridors and occluded shelving zones.

The Meta-RL adaptation engine produced measurable improvements in navigation policy flexibility when robots encountered environmental configurations outside their training distribution. In scenarios involving rearranged shelving units, temporarily blocked corridors, and repositioned floor-level obstacles, the MAML-inspired policy update mechanism enabled rapid reconfiguration of navigation behavior without requiring full retraining cycles. This contrasts directly with standard deep reinforcement learning baselines, which exhibited persistent localization degradation and path planning failures when faced with structural changes to the warehouse layout,

necessitating substantially longer adaptation periods before stable navigation was re-established. The cloud-hosted Meta-RL training loop successfully ingested aggregated sensor and trajectory data from the central server, refined navigation policies, and distributed updated policy parameters back to edge devices within operationally acceptable timeframes. Edge-layer execution of the updated policies preserved real-time obstacle avoidance responsiveness, with local inference running within the latency constraints imposed by the navigation stack.

The object detection pipeline, built around a YOLOv8 model trained exclusively on synthetic data generated within the Gazebo simulation, demonstrated promising generalization to held-out synthetic test configurations. The domain randomization pipeline, which varied lighting conditions, surface textures, and object placements across training scenes, contributed to improved detection robustness in test scenes that differed substantially from any individual training configuration. The trained model approached the target mean Average Precision threshold of mAP@.50 $\geq$ 0.75 on the synthetic evaluation dataset, with the automated annotation pipeline producing pixel-accurate bounding box labels that eliminated manual labelling overhead entirely. The deployed ROS 2 inference node processed camera frames from the simulated robot at a rate meeting the 10 FPS real-time requirement, publishing detection results including class labels, confidence scores, and bounding box coordinates to a dedicated ROS 2 topic for downstream consumption and RViz2 visualization.

The web-based 3D Visualization Platform successfully provided operators with continuous situational awareness throughout multi-robot exploration missions. Live global map updates, robot position overlays, and collision heatmaps rendered responsively through the WebSocket-based data gateway, enabling operators to monitor fleet state without perceptible lag relative to the underlying ROS 2 data streams. The collision detection and heatmap generation module identified and flagged navigation conflict zones in real time, highlighting occupancy grid regions where robot trajectories intersected and where path re-routing interventions were most beneficial. Operators exercising the human-in-the-loop annotation interface marking priority zones, defining keep-out areas, and confirming or rejecting semantically flagged map changes produced measurable refinements in subsequent robot task allocation

decisions, demonstrating that domain-expert interaction with the system translates into improved operational outcomes beyond what fully autonomous decision-making alone achieves.

5.2 Research Finding

The most significant finding of this research is that the integration of SLAM, Meta-RL, and multi-robot collaboration within a unified edge-cloud architecture produces outcomes that none of these components achieves in isolation, directly addressing the Integration Gap identified in the comparative literature analysis. Prior work, including the MAML framework [16] and Zeng et al.'s obstacle avoidance studies [17], demonstrated Meta-RL's potential exclusively in single-robot simulation settings, leaving unanswered how adaptive policy learning behaves when coupled with live multi-robot sensor feedback and collaborative map data. This research finds that the coupling is synergistic: the continuous stream of diverse environmental observations generated by a coordinated robot fleet provides a richer and more varied training signal for Meta-RL policy updates than any single-robot deployment can produce, accelerating policy generalization to unseen configurations. Simultaneously, improved navigation policies fed back to individual robots enhance the quality and spatial coverage of the maps they contribute to the shared global representation, creating a reinforcing cycle between exploration quality and policy improvement that neither component sustains independently.

The decentralized map fusion architecture, operating at a per-robot update frequency of 1 Hz with ICP-based registration and log-odds cell merging, was found to be viable for coordinated teams of two to five robots without overwhelming the central server's processing capacity or degrading the real-time responsiveness of individual robot navigation. This finding addresses the core scalability limitation of centralized SLAM approaches such as RTAB-Map [14], which concentrate all processing burden at a

single node and become bottlenecks as fleet size grows. The namespace-based ROS 2 isolation strategy proved essential to this scalability, preventing TF tree conflicts and topic collisions that would otherwise degrade localization accuracy across the fleet. Importantly, the incremental map registration approach integrating each new robot map against the existing global map rather than re-registering all maps from scratch maintained computational efficiency as the global map grew in spatial extent over extended exploration missions.

The edge-cloud computational split emerged as a functional necessity rather than merely a design preference, representing one of the most operationally significant findings of this work. Feasibility analysis of consolidating all computation on-device, even with GPU-capable embedded hardware such as the NVIDIA Jetson Orin Nano considered during the technical feasibility study, revealed that running real-time SLAM alongside Meta-RL policy inference and object detection simultaneously on a single board introduced navigation latency that degraded performance in dynamically changing environments. By offloading global map fusion, Meta-RL training, and large-scale SLAM optimization to the central server and cloud layer respectively, the edge devices were able to sustain real-time obstacle avoidance and local path planning without computational interference from heavier background tasks. This finding has direct implications for the cost-effective deployment of autonomous navigation platforms, confirming that cloud-assisted architectures enable sophisticated autonomous behavior on hardware that would otherwise be computationally inadequate.

The YOLOv8 object detection pipeline demonstrated that domain randomization applied to synthetic Gazebo training data can produce detection models with sufficient generalization for deployment in varied simulated configurations, providing evidence for the viability of sim-to-real transfer in object recognition tasks for indoor navigation contexts. This finding aligns with and extends prior domain randomization research [22], confirming that randomizing visual properties including lighting, texture, and object placement during synthetic data generation substantially reduces the distribution gap between training and deployment environments. However, the finding also highlighted a boundary condition: while randomization improved robustness

across synthetic test scenes, the degree of visual realism achievable within Gazebo remains a limiting factor for direct real-world transfer without additional fine-tuning on real sensor data. The human-in-the-loop feedback mechanism further revealed an unexpected finding: operator annotations of the semantic map, when incorporated into task allocation decisions, improved fleet coverage efficiency beyond what the autonomous coordination algorithms alone achieved, suggesting that hybrid human-machine decision-making offers performance advantages that fully automated systems cannot replicate.

5.3 Discussion

The results obtained from this platform provide strong support for the central hypothesis motivating this research: that a unified system integrating SLAM-based mapping, Meta-RL adaptive navigation, and decentralized multi-robot collaboration within an edge-cloud architecture can overcome the critical limitations that have prevented any single prior approach from achieving reliable autonomous navigation in dynamic indoor environments. The literature review established that existing contributions including ORB-SLAM [13], MAML [16], CORB-SLAM [18], and offline change detection methods [20][21] address individual dimensions of this problem with considerable sophistication but fail to deliver operationally viable performance when deployed in realistic multi-robot dynamic scenarios, precisely because their architectures were not designed for integration. The findings reported here demonstrate that these components are not only technically compatible but actively enhance each other when embedded in a coherent system: adaptive policies improve map quality, improved maps accelerate policy refinement, and the resulting shared situational awareness enables coordination strategies that single-agent systems cannot support.

The Meta-RL results extend the contribution of Zeng et al. [17] in a particularly meaningful direction. Whereas that work demonstrated the feasibility of Meta-RL for obstacle avoidance in controlled simulation settings, the present platform validates similar adaptation behavior in a multi-robot context where the policy must account for the movements of peer agents as well as static and dynamic environmental features. The MAML-inspired training approach [16] generalizes effectively when the training

signal is enriched by diverse observations from multiple robots exploring different regions of the same environment simultaneously, a configuration that prior single-agent Meta-RL research could not produce. This finding suggests that multi-robot deployment is not merely a scaling challenge for Meta-RL but a qualitative enhancement of the training regime. Similarly, the map fusion results address the centralization limitation identified in CORB-SLAM [18] and the homogeneity assumption of DDF-SLAM [19] by demonstrating that incremental ICP-based fusion with probabilistic cell merging, built on established factor graph foundations [24], can produce globally consistent maps from heterogeneous robot observations without requiring network-synchronous communication or uniform hardware specifications.

Several important limitations were encountered during development and evaluation that qualify the generalizability of these findings. Network reliability between edge devices and the central server proved to be the most operationally significant vulnerability: under simulated conditions of elevated interference or bandwidth congestion, the quality of navigation decisions during connectivity disruptions was noticeably reduced compared to cloud-assisted operation, even with local policy caching active as a fallback mechanism. This finding indicates that offline contingency modes cannot be treated as secondary features in future iterations; they must be designed as primary operational modes with equivalent performance targets to connected operation. A second limitation arose from hardware calibration inconsistencies between robot units during heterogeneous team configurations: minor differences in LiDAR scan resolution and mounting offsets between prototype units introduced map fusion artefacts that required additional post-processing steps not anticipated in the original architecture. While the modular sensor abstraction design mitigated the most severe misalignments, a more robust calibration normalization layer will be required before the framework can be deployed on fully heterogeneous fleets without manual pre-calibration.

The ICP registration process also revealed a performance boundary specific to environments with highly repetitive structural features, such as the parallel shelving aisles in the Gazebo warehouse configuration. In these regions, the ICP algorithm's iterative nearest-neighbor matching converged to locally optimal but geometrically

incorrect alignments due to the ambiguity introduced by repeated structural patterns, producing minor but cumulative map drift over extended exploration sessions. This observation highlights a known limitation of geometry-only registration approaches and motivates the integration of semantic landmarks or place recognition features such as those explored by Sünderhauf et al. [20] as disambiguation anchors in future iterations of the fusion pipeline. Addressing this limitation would simultaneously improve map fusion robustness in repetitive environments and advance the semantic-aware change detection capabilities of the platform, as semantically labelled landmarks provide stable reference points against which structural changes can be detected and versioned.

The broader implications of this platform extend across several high-value application domains. In warehouse automation, the combination of real-time adaptive navigation and collaborative exploration directly addresses the operational challenges posed by dynamic inventory environments where obstacle configurations change continuously with logistics activity. In hospital settings, where patient and staff movement patterns are unpredictable and safety requirements are stringent, the human-in-the-loop intervention mechanism provides the operational oversight necessary for responsible autonomous deployment. In disaster response scenarios, where communication infrastructure may be degraded and environmental conditions are unknown and hazardous, the edge-cloud architecture's ability to sustain local autonomous operation during connectivity disruptions and the Meta-RL engine's capacity for rapid policy adaptation to previously unseen environments are particularly well-matched to operational requirements. Future work should prioritize the development of robust offline contingency modes, more sophisticated sensor abstraction layers for heterogeneous fleet support, federated Meta-RL training strategies that protect sensitive operational data while enabling policy sharing across deployed fleets, and real-world validation trials in each of these application domains to assess performance under conditions that extend beyond the controlled simulation baseline established in this research.

6 SUMMARY OF STUDENT CONTRIBUTION

This research represents an original and integrated contribution to the field of autonomous indoor navigation, addressing a gap that the literature review clearly identified: the absence of a unified platform that brings together SLAM-based mapping, Meta-RL adaptive navigation, decentralized multi-robot coordination, and real-time visualization within a single operational framework.The division of responsibilities was aligned with the four core components of the system architecture: the Meta-RL adaptive navigation engine, the 3D Mapping and Visualization Platform, the Multi-Robot Collaboration Framework, and the Simulation Platform. While each contributor maintained primary ownership of their respective component, the interdependent nature of the platform required continuous coordination across all four workstreams to ensure architectural compatibility, data format consistency, and end-to-end system integration.

Thilochana J M was responsible for the design, development, and validation of the Meta-RL adaptive navigation engine. This contribution encompassed the implementation of the MAML-inspired policy training pipeline, which leverages aggregated multi-robot sensor data transmitted from the central server to iteratively refine navigation strategies in response to previously unseen environmental configurations. Thilochana engineered the cloud-hosted training loop that ingests trajectory and sensor observations, optimizes navigation policies, and distributes updated policy parameters back to edge devices via the deployment platform. The prototype robot integrating an Arduino-based control system with cloud-assisted inference was assembled and validated as the physical testbed for confirming that effective adaptive navigation is achievable on cost-constrained hardware when heavy computation is offloaded to the cloud layer. Additionally, Thilochana developed the real-time monitoring and performance evaluation interface, which provides operators with live visibility into robot locations, navigation metrics, and system health parameters, and incorporates human feedback into ongoing policy refinement cycles. This contribution directly addresses the research gap identified in the literature concerning the simulation-only limitations of prior Meta-RL approaches [16][17],

extending adaptive policy learning into a real-world, multi-robot operational context for the first time within this platform.

Jayarathne H C D was responsible for the design and implementation of the 3D Mapping and Visualization Platform, which serves as the central hub for environmental state management and human operational oversight across the entire multi-robot fleet. This contribution involved the development of a modular SLAM backend capable of receiving, fusing, and versioning partial maps from multiple robots into a consistent global 3D representation, supported by a change detection pipeline that continuously compares incoming map data against stored versions to identify and timestamp structural environmental modifications. Jayarathne designed and built the interactive web-based 3D visualization client using modern rendering technologies, providing operators with real-time views of the evolving global map, live robot position overlays, and collision heatmap outputs generated from historical trajectory data. The low-latency WebSocket data synchronization gateway, which streams map updates, robot poses, and system alerts from the backend to the browser client with minimal perceptible delay, was engineered as part of this contribution. Advanced diagnostic tools including the version-controlled map diff viewer and the user-interactive annotation system enabling operators to tag obstacles, define exclusion zones, and confirm or reject semantically detected changes were also developed within this workstream. This component directly addresses the visualization and diagnostic gap identified in the background survey [12], providing the operational transparency required for effective human oversight of a complex autonomous fleet.

Senanayake S M A S N was responsible for the design, implementation, and testing of the Multi-Robot Collaboration Framework, which constitutes the coordination backbone of the platform and enables teams of two to five heterogeneous robots to explore, map, and navigate shared indoor environments in a decentralized yet coherent manner. This contribution encompassed the implementation of the ROS 2 topic-based communication infrastructure, including namespace-based isolation of each robot's data streams to prevent TF tree conflicts and topic collisions across the fleet, and the configuration of SLAM Toolbox instances operating in asynchronous mode on each robot to ensure that map-building operations do not introduce latency into real-time

navigation cycles. Senanayake developed the Map Fusion Engine, implementing TF coordinate frame alignment, ICP-based map registration to correct for odometry drift and localization discrepancies, and a log-odds probabilistic cell merging strategy that reconciles conflicting sensor readings from multiple agents into a unified global occupancy grid updated incrementally in real time. The Collision Detection and Heatmap Generation Module, which analyzes the merged global map to identify high-density obstacle regions and generate trajectory heatmaps highlighting navigation conflict zones, was also implemented within this workstream. Dynamic task coordination mechanisms, including shared costmap maintenance and path conflict detection, were developed to minimize redundant coverage and maximize exploration efficiency across the fleet. This contribution addresses the centralized control and homogeneity limitations of prior multi-robot SLAM frameworks [18][19], demonstrating that decentralized map fusion with probabilistic merging is viable and scalable for heterogeneous robot teams.

GOMIS R J S was responsible for the design and implementation of the Simulation Platform, which provides the foundational virtual environment underpinning the data generation, algorithm development, and system validation activities of the entire project. This contribution involved the configuration of a high-fidelity Gazebo warehouse simulation environment incorporating static structural elements shelving units, perimeter walls, loading bays, and floor-level obstacles that present the multi-robot system with a representative and controlled exploration challenge. GOMIS developed the automated synthetic data generation pipeline, implementing scenario controller scripts that systematically reposition the simulated robot across diverse viewpoints, a domain randomizer that procedurally varies lighting conditions, surface textures, and object placements across training scenes, and a data collector and annotator that captures camera imagery alongside pixel-accurate bounding box labels in YOLO annotation format. The YOLOv8 object detection model was trained on this synthetic dataset, with hyperparameter tuning and data augmentation strategies applied to maximize generalization, and the trained model was deployed as a ROS 2 inference node capable of processing simulated camera feeds at real-time frame rates and publishing detection results including class labels, confidence scores, and bounding

box coordinates to a dedicated ROS 2 topic for visualization in RViz2. The simulation environment additionally served as the primary testbed for validating the Multi-Robot Collaboration Framework and Meta-RL training pipeline prior to real-world prototype trials, ensuring that all components were exercised under controlled and reproducible conditions before physical deployment. This contribution provides the synthetic data infrastructure and virtual validation environment upon which the reliability of the broader platform's claims rests, building on domain randomization principles established in prior sim-to-real transfer research [22][23].

Collectively, the four contributions form an end-to-end autonomous navigation platform that bridges the gap between isolated academic research and practical multi-robot deployment. The integration of these four workstreams adaptive navigation, environmental visualization, multi-robot coordination, and simulation-based validation delivers the unified system architecture that the research problem demanded, with each component designed for independent upgradeability to accommodate future advances in algorithms, sensors, and robotic platforms. Collectively, these contributions bridge the divide between isolated academic research and practical deployment, providing the robotics community with an end-to-end reference architecture for scalable, adaptive, and collaborative indoor navigation. The modular design of the platform ensures that individual components can be independently upgraded, making the system extensible as new algorithms, sensors, and robotic platforms emerge.

7 CONCLUSION

This research set out to address a persistent and well-documented gap in autonomous robotics: the absence of a comprehensive platform that unifies Simultaneous Localization and Mapping, Meta-Reinforcement Learning, and multi-robot collaboration into a single, operationally viable system. The work presented in this proposal and its subsequent development demonstrates that such integration is not only

technically feasible but practically necessary for autonomous mobile robots to perform reliably in the dynamic, unstructured indoor environments that real-world applications demand.

The platform developed through this research brings together four interdependent capabilities that prior literature had largely treated in isolation. The SLAM backbone provides continuous, high-fidelity environmental mapping that adapts to structural changes over time. The Meta-RL engine enables navigation policies to be updated rapidly in response to previously unseen conditions, overcoming the fundamental limitation of traditional reinforcement learning approaches that require extensive retraining when environments shift. The multi-robot collaboration framework enables teams of heterogeneous robots to share situational awareness, coordinate exploration tasks, and resolve navigational conflicts in a decentralized manner that scales beyond what any single-agent system can achieve. Finally, the edge-cloud architecture distributes computational responsibilities in a way that preserves real-time responsiveness at the robot level while making heavy processing tasks such as global map fusion and policy optimization tractable through cloud infrastructure.

The findings confirmed that integrating these components produces outcomes that none of them achieves independently. Collaborative exploration reduced redundancy and improved area coverage efficiency. Meta-RL adaptation maintained navigation performance in dynamic conditions where conventional algorithms exhibited localization degradation. The edge-cloud split enabled cost-effective deployment on lightweight hardware without sacrificing the computational depth required for sophisticated decision-making. The web-based visualization and human-in-the-loop feedback mechanism further ensured that the system remained interpretable and correctable by operators, an essential quality for deployment in safety-critical environments such as hospitals, warehouses, and disaster response scenarios.

Several limitations identified during this work also point toward productive directions for future research. Network reliability under congestion remains a vulnerability that warrants the development of more capable offline operational modes. Sensor heterogeneity across robot teams introduces calibration challenges that call for more

sophisticated data abstraction layers. The computational cost of simultaneous real-time SLAM and policy inference at the edge suggests that further investigation into lightweight model compression and edge-optimized neural architectures would be valuable.

Looking forward, the modular architecture of this platform is deliberately designed to accommodate these improvements without requiring fundamental redesign. New SLAM backends, updated Meta-RL training strategies, additional robot types, and expanded sensor modalities can each be integrated independently, making the system extensible alongside advances in the broader robotics and machine learning fields. The platform therefore serves not only as a solution to the specific problem addressed in this research but as a reusable foundation upon which future work in autonomous multi-robot navigation can build.

In conclusion, this research demonstrates that the convergence of SLAM, Meta-RL, and collaborative robotics within a well-engineered software and hardware architecture can significantly advance the state of the art in autonomous indoor navigation. By bridging the gap between isolated academic contributions and integrated real-world deployment, this work contributes a meaningful step toward robotic systems that are adaptive, collaborative, and robust enough to operate reliably in the complex and ever-changing environments that characterize the spaces humans inhabit and work in every day.

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9 APPENDIX